

Sharp pointwise lower bounds for Lebesgue functions

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Abstract

For each n , let λ_n be the Lebesgue function for polynomial interpolation at an arbitrary set of n distinct nodes in $[-1, 1]$. We prove that the points $x \in (-1, 1)$ for which $\lambda_n(x) > (2/\pi) \log n - C(x)$ for infinitely many n , with some finite constant $C(x)$, form a dense set. We also prove that $\limsup_{n \rightarrow \infty} \lambda_n(x)/\log n \geq 2/\pi$ for almost every $x \in (-1, 1)$. The constant $C(x)$ may depend on the array and on x . The first proof rests on an energy estimate for the derivative jumps of λ_n at the nodes. The second proof uses positive Cauchy transforms and harmonic measure and is independent of Tao's local Bernstein theory.

1 Introduction

For each $n \geq 1$, let

$$-1 \leq x_{1,n} < \cdots < x_{n,n} \leq 1.$$

Define the node polynomial, the cardinal polynomials, and the Lebesgue function by

$$P_n(x) = \prod_{k=1}^n (x - x_{k,n}),$$
$$\ell_{k,n}(x) = \frac{P_n(x)}{(x - x_{k,n})P'_n(x_{k,n})}, \quad \lambda_n(x) = \sum_{k=1}^n |\ell_{k,n}(x)|.$$

The cardinal polynomials are interpreted by continuity at the nodes. The rows of the array need not be nested.

Theorem 1. *For every triangular array of distinct nodes as above, the following statements hold.*

(i) *The set of points $x \in (-1, 1)$ for which there is a finite constant $C(x)$ such that*

$$\lambda_n(x) > \frac{2}{\pi} \log n - C(x) \tag{1.1}$$

for infinitely many n is dense in $(-1, 1)$.

(ii) *For Lebesgue almost every $x \in (-1, 1)$,*

$$\limsup_{n \rightarrow \infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}. \tag{1.2}$$

Erdős posed the existence and almost-everywhere questions for triangular arrays in [4, p. 68], and they are recorded as Problem 1132 in [2]. The constant in (1.1) is independent of n ; it may depend on the array and the point x .

Faber [8] proved the order $\log n$ for the global Lebesgue constant, Bernstein [1] obtained the asymptotically sharp leading constant $2/\pi$, Erdős and Turán [6] improved the error to $O(\log \log n)$, Erdős [5] to $O(1)$, and Vértési [10, Corollary 1] refined the asymptotics through the optimal constant term to an error of $O((\log \log n / \log n)^2)$. See [9, Section 1.3] and [3] for further historical discussion.

For triangular arrays, Bernstein's result [1] implies that the set of points x satisfying

$$\limsup_{n \rightarrow \infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}$$

is everywhere dense; Erdős explicitly asked whether this holds for almost every x [4, p. 68]. Erdős and Vértési [7] proved that for every triangular array there is a continuous function whose interpolants diverge almost everywhere; in particular, $\limsup_{n \rightarrow \infty} \lambda_n(x) = \infty$ for almost every x . Thus Theorem 1(ii) gives the sharp logarithmic rate almost everywhere.

Tao [9, Theorem 1.10(i)] proved that every fixed nondegenerate interval $I \subset [-1, 1]$ satisfies

$$\sup_{x \in I} \lambda_n(x) \geq \frac{2}{\pi} \log n - O_I(1).$$

His Corollary 1.11 and Remark 1.12 give, for every prescribed $\omega(n) \rightarrow \infty$, a dense, indeed comeager, set of points x for which

$$\lambda_n(x) \geq \frac{2}{\pi} \log n - \omega(n)$$

infinitely often. Theorem 1(i) replaces $\omega(n)$ by a finite constant depending on the point, on a dense set. Baire category enters (i) at one point, to produce an interval with a uniform upper bound; recurrence on a positive-measure set follows from the energy estimate in Section 3.

Throughout the paper, all logarithms are natural and $|E|$ denotes Lebesgue measure. We write $c_0 = 2/\pi$, and suppress the row index when discussing a single row. Constants in local estimates may depend on the fixed intervals and test functions, but not on n or on interpolation data bounded by one. The notation $J \Subset U$ means that J is a compact subset of the open set U .

2 A local differentiation formula

We use the following local form of Riesz's differentiation formula and its near-equality consequence.

Lemma 2. *Suppose that F is holomorphic in a neighborhood of the square $\{|\Re z|, |\Im z| \leq T\}$, is real on its real diameter, and satisfies, throughout this square,*

$$|F(u + iv)| \leq e^{|v|}.$$

Then

$$|F'(0)| \leq 1 + O(T^{-1}). \tag{2.1}$$

If $F'(0) \geq 1 - \delta$, where $\delta \geq 0$, then

$$F(\pi/2) \geq 1 - \frac{\pi^2}{4} \delta - O(T^{-1}). \tag{2.2}$$

The constants are absolute.

Proof. Choose $R = m\pi$ with $T/2 \leq R < T$, and integrate

$$\frac{F(z)}{z^2 \cos z}$$

around the square of radius R . On its vertical sides, $|\cos z| = \cosh(\Im z)$; on its horizontal sides, $|\cos z| \geq \sinh R$. The contour integral, divided by $2\pi i$, is consequently $O(R^{-1})$. Taking residues at zero and at $t_j = (j + \frac{1}{2})\pi$ gives

$$F'(0) = \sum_{|t_j| < R} \frac{(-1)^j F(t_j)}{t_j^2} + O(T^{-1}). \quad (2.3)$$

Since $|F(t_j)| \leq 1$ and

$$\sum_{j \in \mathbb{Z}} \frac{1}{t_j^2} = 1,$$

equation (2.1) follows. Isolating the term $t_0 = \pi/2$ in (2.3) gives

$$1 - \delta \leq \frac{4}{\pi^2} F(\pi/2) + 1 - \frac{4}{\pi^2} + O(T^{-1}),$$

which proves (2.2). $\square$

We apply this formula to interpolants with bounded nodal data. The estimates we use from Tao are stated first, with their normalization:

$$S_n = \sum_k \frac{1}{|P'_n(x_{k,n})|}, \quad \alpha_n = \frac{1}{n} \log S_n, \quad U_n(z) = \frac{1}{n} \log \frac{1}{|P_n(z)|}.$$

Proposition (Tao's local estimates). *Let $I \subseteq (-1, 1)$ be a fixed nondegenerate closed interval, and suppose that $\sup_I \lambda_n \ll n^{O(1)}$. With α_n normalized as above [9, equation (4.11)], there are densities ρ_n on $\text{int } I$ with the following properties. For each fixed compact interval $I' \subseteq \text{int } I$:*

1. *The densities are bounded above and below by positive constants and have uniformly bounded Lipschitz norms:*

$$\rho_n(x) \asymp_{I'} 1, \quad |\rho_n(x) - \rho_n(x_0)| \ll_{I'} |x - x_0| \quad (x, x_0 \in I').$$

This is [9, Theorem 4.1(ii), equation (4.6)].

2. *For $x \in I'$ and $\eta \geq \log n/n$,*

$$U_n(x + i\eta) = \alpha_n - \pi\eta\rho_n(x) + O_{I'}\left(\frac{\log n}{n} + \eta^3\right). \quad (2.4)$$

This is [9, equation (4.7)].

3. *For every subinterval $K \subset I'$ and every $A \geq 1$,*

$$\frac{1}{n} \#\{k : x_{k,n} \in K\} = \int_K \rho_n(x) dx + O_{I'}\left(\frac{|K|}{A} + \frac{A^2 \log n}{n}\right). \quad (2.5)$$

This is [9, Theorem 4.1(iv), equation (4.9)].

The function ρ_n here is Tao's density defined from the original interval I by his equation (4.15), so it does not depend on the auxiliary choice of I' . Taking $A = n^{1/10}$ in (2.5) and integrating by parts shows that, for functions h_n supported in a fixed compact subset of $\text{int } I$ with uniformly bounded Lipschitz norms,

$$\frac{1}{n} \sum_k h_n(x_{k,n}) = \int h_n(x) \rho_n(x) dx + O(n^{-1/10}). \quad (2.6)$$

Lemma 3. *Let $I \subseteq (-1, 1)$ be a fixed nondegenerate closed interval. Suppose that, for all sufficiently large n ,*

$$\lambda_n(x) \leq \Lambda_n := c_0 \log n + C \quad (x \in I), \quad (2.7)$$

where C is fixed. This bound gives the hypothesis of the proposition above; let ρ_n be its densities. For every fixed interval $J \subseteq \text{int } I$ and every real interpolant

$$q(x) = \sum_{k=1}^n a_k \ell_{k,n}(x), \quad |a_k| \leq 1,$$

uniformly for $x \in J$,

$$|q'(x)| \leq \pi n \rho_n(x) (\Lambda_n + o(1)), \quad |q''(x)| \leq C_J n^2 \log n. \quad (2.8)$$

Moreover, for every fixed D , if

$$\frac{|q'(x)|}{\pi n \rho_n(x)} \geq c_0 \log n - D, \quad (2.9)$$

then there is $y = x + O_J(n^{-1})$ such that

$$\lambda_n(y) \geq c_0 \log n - D', \quad (2.10)$$

where D' depends on I, J, C, D , but not on n, x, q . The point y lies in any fixed compact interior interval containing J in its interior, for all sufficiently large n .

Proof. Fix $I' \subseteq \text{int } I$ with $J \subseteq \text{int } I'$. For $\Re z \in I'$ and $\Im z = \eta \geq \log n/n$, the interpolation formula and (2.4) give

$$|q(z)| \leq \frac{S_n |P_n(z)|}{\eta} \leq \frac{1}{\eta} \exp\left(\pi n \eta \rho_n(\Re z) + O(\log n + n \eta^3)\right). \quad (2.11)$$

Fix $x \in J$, and put

$$L = n^{-1/5}, \quad H = n^{-2/5}, \quad \omega = \pi n \rho_n(x) (1 + n^{-1/10}).$$

On the upper edge of $\{|\Re z - x| \leq L, 0 \leq \Im z \leq H\}$, we have $|q(z)| \leq \Lambda_n e^{\omega H}$: ωH exceeds $\pi n H \rho_n(x)$ by $\pi \rho_n(x) n^{1/2}$, which dominates the $O(n^{2/5})$ from the Lipschitz variation of ρ_n and the remaining $O(\log n)$ terms in (2.11). On the lower edge $|q| \leq \Lambda_n$; a Chebyshev expansion on I gives $|q| \leq \Lambda_n \exp(O_I(n))$ on the vertical sides. Apply [9, Lemma 2.5] to $\Lambda_n^{-1} q(z) e^{i\omega(z-x)}$. At distance at least $L/2$ from the vertical sides, its error factor is $\exp(O(ne^{-cL/H})) = 1 + O(n^{-10})$. Thus, also reflecting across the real axis,

$$|q(x + u + iv)| \leq \Lambda_n (1 + n^{-10}) e^{\omega|v|} \quad (|u| \leq L/2, |v| \leq H). \quad (2.12)$$

The rescaled function

$$F(t) = \frac{q(x + t/\omega)}{\Lambda_n (1 + n^{-10})}$$

satisfies Lemma 2 on a square of radius $T \asymp_J n^{3/5}$. By (2.1),

$$|q'(x)| = \omega \Lambda_n (1 + n^{-10}) |F'(0)| \leq \pi n \rho_n(x) \Lambda_n (1 + n^{-1/10}) (1 + O(n^{-3/5})).$$

Since $\Lambda_n n^{-1/10} = o(1)$, this gives the first bound in (2.8); Cauchy's estimate on a disk of radius comparable to n^{-1} gives the second. Under (2.9), change the sign of q if necessary. For sufficiently large n ,

$$F'(0) \geq \frac{c_0 \log n - D}{\Lambda_n (1 + n^{-1/10}) (1 + n^{-10})} \geq 1 - \frac{C_1}{\Lambda_n}, \quad C_1 = 1 + \max\{C + D, 0\}.$$

Equation (2.2) then gives $|q(x + \pi/(2\omega))| \geq \Lambda_n - C_2$. Since $|q(y)| \leq \lambda_n(y)$ on the real axis and $\omega \asymp_J n$, this proves (2.10) with the asserted displacement. $\square$

3 Derivative jumps and a Riesz energy

At each internal interpolation node, write

$$J_k = \lambda'_n(x_{k,n}+) - \lambda'_n(x_{k,n}-).$$

At the first and last nodes, use the exterior polynomial pieces. Set

$$\nu_k = \frac{1}{S_n |P'_n(x_{k,n})|}.$$

The absolute values of all cardinal polynomials other than $\ell_{k,n}$ have a cusp at $x_{k,n}$, whereas $\ell_{k,n}$ is positive near that node. Using $\ell_{j,n}(x) = P_n(x)/((x - x_{j,n})P'_n(x_{j,n}))$ and $P_n(x_{k,n}) = 0$ to differentiate at $x_{k,n}$, we obtain

$$J_k = 2 \sum_{j \neq k} |\ell'_{j,n}(x_{k,n})| = 2 \sum_{j \neq k} \frac{\nu_j}{\nu_k |x_{k,n} - x_{j,n}|}. \quad (3.1)$$

Lemma 4. *Under the hypotheses of Lemma 3, choose a nonzero nonnegative smooth function f compactly supported in $\text{int } I$. Then*

$$\sum_k \frac{f(x_{k,n})^2}{n \rho_n(x_{k,n})} \frac{J_k}{2\pi n \rho_n(x_{k,n})} \geq c_0 \log n \int f(x)^2 dx - O(1). \quad (3.2)$$

Terms with $f(x_{k,n}) = 0$ are taken to be zero; ρ_n is only needed on a fixed interior neighborhood of the support of f .

Proof. Put $a_k = f(x_{k,n})/\rho_n(x_{k,n})$, with $a_k = 0$ outside the support of f , and define

$$\mu_n = \frac{1}{n} \sum_k a_k \delta_{x_{k,n}}, \quad K_\varepsilon(t) = \frac{1}{\sqrt{t^2 + \varepsilon^2}}.$$

The kernel K_ε is positive definite, since

$$K_\varepsilon(t) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-s\varepsilon^2} e^{-st^2} ds$$

is a positive mixture of Gaussian kernels. Write

$$\mathcal{E}_\varepsilon(\sigma, \tau) = \iint K_\varepsilon(x - y) d\sigma(x) d\tau(y)$$

for its bilinear energy. Positivity applied to $\mu_n - f(x) dx$ gives

$$\mathcal{E}_\varepsilon(\mu_n, \mu_n) \geq 2\mathcal{E}_\varepsilon(\mu_n, f dx) - \mathcal{E}_\varepsilon(f dx, f dx). \quad (3.3)$$

Take $\varepsilon = 1/n$. Subtracting $f(x)$ in the integral over $|t| \leq 1$, and using the Lipschitz bound for f , gives the uniform estimate

$$(K_{1/n} * f)(x) = 2 \log n f(x) + O_f(1). \quad (3.4)$$

The mass of μ_n is bounded. Also, by (2.6),

$$\int f d\mu_n = \frac{1}{n} \sum_k \frac{f(x_{k,n})^2}{\rho_n(x_{k,n})} = \int f^2 + O(n^{-1/10}). \quad (3.5)$$

Again by (3.4),

$$\mathcal{E}_{1/n}(f dx, f dx) = 2 \log n \int f^2 + O(1).$$

Together with (3.3)–(3.5), this yields

$$\mathcal{E}_{1/n}(\mu_n, \mu_n) \geq 2 \log n \int f^2 - O(1). \quad (3.6)$$

The diagonal part of this energy is $n^{-1} \sum_k a_k^2 = O(1)$. Since $K_{1/n}(t) \leq 1/|t|$ for $t \neq 0$,

$$\frac{1}{n^2} \sum_{k \neq j} \frac{a_k a_j}{|x_{k,n} - x_{j,n}|} \geq 2 \log n \int f^2 - O(1). \quad (3.7)$$

Finally,

$$a_k^2 \frac{\nu_j}{\nu_k} + a_j^2 \frac{\nu_k}{\nu_j} \geq 2a_k a_j.$$

By (3.1), the left side of (3.2) is

$$\frac{1}{\pi n^2} \sum_{k \neq j} \frac{a_k^2 \nu_j}{\nu_k |x_{k,n} - x_{j,n}|}.$$

Pairing the terms indexed by (k, j) and (j, k) , and then applying (3.7), bounds this expression below by

$$\frac{1}{\pi n^2} \sum_{k \neq j} \frac{a_k a_j}{|x_{k,n} - x_{j,n}|} \geq \frac{2}{\pi} \log n \int f^2 - O(1).$$

This proves (3.2). □

4 A dense set of points with bounded additive loss

Proposition 5. *Assume the local upper bound (2.7) on a fixed nondegenerate interval $I \Subset (-1, 1)$. There are a constant D and a measurable set $G \subset \text{int } I$, with $|G| > 0$, such that every $x \in G$ satisfies*

$$\lambda_n(x) \geq c_0 \log n - D$$

for infinitely many n .

Proof. Choose compact intervals $J_0 \Subset J_1 \Subset \text{int } I$, and choose f as in Lemma 4 with $\text{supp } f \subset J_0$. All constants below are independent of n : after I, C, J_0, J_1, f are fixed, we choose D , then $\kappa, D_1, d, \kappa_1, d_1$, and finally b, D_2 . Let

$$w_k = \frac{f(x_{k,n})^2}{n\rho_n(x_{k,n})}, \quad T_k = \frac{J_k}{2\pi n\rho_n(x_{k,n})}, \quad W = \int f^2 > 0.$$

By (2.6),

$$\sum_k w_k = W + O(n^{-1/10}), \quad 0 \leq w_k \leq C_f/n. \quad (4.1)$$

On each interval between successive nodes, λ_n equals an interpolant with nodal data in $\{-1, 1\}$. Both one-sided derivatives at a node therefore satisfy (2.8), so

$$0 \leq T_k \leq \Lambda_n + o(1) \quad (4.2)$$

uniformly on the support of f . Lemma 4, (4.1), and (4.2) show that

$$\sum_k w_k(\Lambda_n + 1 - T_k) = O(1), \quad (4.3)$$

with nonnegative summands for all sufficiently large n . On a node with $T_k < c_0 \log n - D$, the factor in (4.3) is greater than $C + 1 + D$. Hence, by choosing D sufficiently large, such nodes carry at most $W/2$ of the weights. Since (4.1) has total weight $W + o(1)$, for all sufficiently large n the nodes in the support of f satisfying

$$T_k \geq c_0 \log n - D \quad (4.4)$$

carry at least $W/4$ of the weights. In particular, there are at least κn such nodes, with $\kappa > 0$ independent of n . Discarding the first and last nodes does not change this conclusion after reducing κ .

At a node satisfying (4.4), at least one adjacent polynomial piece q has

$$|q'(x_{k,n})| \geq \pi n \rho_n(x_{k,n})(c_0 \log n - D).$$

Lemma 3 gives a point

$$y_k = x_{k,n} + O(n^{-1}), \quad \lambda_n(y_k) \geq c_0 \log n - D_1. \quad (4.5)$$

The same polynomial piece q has value 1 at both endpoints of the corresponding gap. Rolle's theorem and (2.8) imply that this gap has length at least d/n for some fixed $d > 0$ when the gap is contained in J_1 . If it is not contained in J_1 , its length is bounded below by $\text{dist}(J_0, \mathbb{R} \setminus J_1) > 0$.

An interval of length less than d/n contains at most two such nodes, so every interval of length $O(n^{-1})$ contains only a bounded number of the nodes in (4.4). By (4.5), the same holds, with multiplicities, for the points y_k . Hence one can select at least $\kappa_1 n$ of them, separated by d_1/n , for fixed $\kappa_1, d_1 > 0$.

Equation (2.8), applied to every polynomial piece, also shows that λ_n is $C_I n \log n$ -Lipschitz on a fixed compact interior interval containing these points. Choose a fixed sufficiently small $b > 0$, and let A_n be the union of the intervals of radius

$$r_n = \frac{b}{n \log n}$$

centered at the selected points. For all sufficiently large n , these intervals are disjoint, lie in a fixed compact interior interval, and satisfy

$$\lambda_n(x) \geq c_0 \log n - D_2 \quad (x \in A_n), \quad \frac{c}{\log n} \leq |A_n| \leq \frac{C_2}{\log n}. \quad (4.6)$$

The separation of the centers in the m -th row implies, for every interval B ,

$$|A_m \cap B| \leq C_3 \left(\frac{|B|}{\log m} + \frac{1}{m \log m} \right).$$

Summing this inequality over the at most n components of A_n gives, for $m > n$,

$$|A_n \cap A_m| \leq C_3 \left(\frac{|A_n|}{\log m} + \frac{n}{m \log m} \right). \quad (4.7)$$

Put $B_j = A_{2^j}$. Equations (4.6) and (4.7) give

$$|B_j| \asymp j^{-1}, \quad |B_j \cap B_k| \leq C_4 \left(\frac{1}{jk} + \frac{2^{j-k}}{k} \right) \quad (j < k). \quad (4.8)$$

For every fixed j_0 , the sum of the measures of B_j , $j_0 \leq j \leq N$, grows at least as a positive constant times $\log N$. By (4.8), the integral of the square of the sum of their indicators is $O((\log N)^2)$. Cauchy–Schwarz consequently gives a positive lower bound, independent of j_0 , for

$$\left| \bigcup_{j \geq j_0} B_j \right|.$$

Continuity of measure along these decreasing tail unions shows that

$$|\limsup_{j \rightarrow \infty} B_j| > 0. \quad (4.9)$$

Together with (4.6), this proves the proposition. $\square$

Proof of Theorem 1(i). Fix a nondegenerate compact interval $K_0 \subset (-1, 1)$, and suppose that it contains no point satisfying the assertion. Then

$$\lambda_n(x) - c_0 \log n \longrightarrow -\infty \quad \text{for every } x \in K_0. \quad (4.10)$$

The closed sets

$$F_M = \{x \in K_0 : \lambda_n(x) \leq c_0 \log n + M \text{ for every } n \geq 2\}, \quad M = 1, 2, \dots,$$

cover K_0 , by (4.10). Baire's theorem supplies a nondegenerate closed subinterval $I \subset K_0$ on which (2.7) holds with a fixed C . Proposition 5 contradicts (4.10). Thus every such K_0 contains a good point, proving density. Increasing the final additive constant by one gives the strict inequality in the statement. $\square$

5 Positive Cauchy transforms

Suppress the row index and put

$$S = \sum_k \frac{1}{|P'(x_k)|}, \quad \nu_k = \frac{1}{S|P'(x_k)|}, \quad \epsilon_k = \operatorname{sgn} P'(x_k).$$

Define

$$m(z) = \sum_k \frac{\nu_k}{z - x_k}, \quad g(z) = \sum_k \frac{\epsilon_k \nu_k}{z - x_k} = \frac{1}{SP(z)}. \quad (5.1)$$

The last identity is the partial fraction formula for $1/P$, whose expansion at infinity gives

$$\sum_k \epsilon_k \nu_k x_k^j = 0 \quad (0 \leq j \leq n-2). \quad (5.2)$$

For $n \geq 2$, both $m + g$ and $m - g$ are Cauchy transforms of positive probability measures: their masses at x_k are $(1 \pm \epsilon_k) \nu_k$.

For real x away from the nodes, write

$$H(x) = \sum_k \frac{\nu_k}{|x - x_k|}, \quad \gamma_s(x) = -\Im m(x + is), \quad p_s(t) = \frac{s}{\pi(t^2 + s^2)}.$$

Then

$$\lambda_n(x) = \frac{H(x)}{|g(x)|}, \quad H(x) = \frac{2}{\pi} \int_0^\infty \gamma_s(x) \frac{ds}{s}, \quad (5.3)$$

and the Poisson semigroup gives

$$\gamma_s = p_{s-h} * \gamma_h \quad (s > h). \quad (5.4)$$

Lemma 6. *For every row with $n \geq 2$, every $0 < h \leq 1$, and every $x \in [-1, 1]$,*

$$\frac{|g(x + ih)|}{\gamma_h(x)} \leq \frac{4 + h^2}{h^2 \sinh((n-1)h/2)}. \quad (5.5)$$

Consequently, if $0 < a < 1$ and $h = n^{-a}$, then

$$\sup_{x \in [-1, 1]} \frac{|g(x + ih)|}{\gamma_h(x)} \longrightarrow 0. \quad (5.6)$$

Proof. Set $z = x + ih$ and $p = T_{n-1}$, the Chebyshev polynomial of the first kind. The polynomial $(p(t) - p(z))/(t - z)$ has degree at most $n-2$, so (5.2) gives

$$p(z)g(z) = \sum_k \frac{\epsilon_k \nu_k p(x_k)}{z - x_k}, \quad |p(z)g(z)| \leq \frac{1}{h},$$

using $|p(x_k)| \leq 1$ and $\sum_k \nu_k = 1$. Write $z = \cos(u + iv)$, with $u, v \in \mathbb{R}$. Then $h \leq \sinh |v|$, so $|v| \geq \operatorname{arsinh} h \geq h/2$. The identity $T_{n-1}(\cos w) = \cos((n-1)w)$ yields

$$|p(z)|^2 = \cos^2((n-1)u) + \sinh^2((n-1)v) \geq \sinh^2((n-1)h/2).$$

Finally, since the nodes and x lie in $[-1, 1]$,

$$\frac{h}{4 + h^2} \leq \gamma_h(x) = \sum_k \frac{\nu_k h}{(x - x_k)^2 + h^2}.$$

Combining these bounds proves (5.5). For $h = n^{-a}$, its right-hand side tends to zero: the hyperbolic sine grows exponentially in n^{1-a} , while h^{-2} grows polynomially. This proves (5.6). $\square$

Lemma 7. *Let F be the Cauchy transform of a positive probability measure with finite support. For every bounded real interval S_0 , every $x \in \mathbb{R}$, and every $h > 0$,*

$$\int_{\mathbb{R}} p_h(x - y) \mathbf{1}_{\{F(y) \in S_0\}} dy = \frac{1}{\pi} \int_{S_0} \frac{-\Im F(x + ih)}{(t - \Re F(x + ih))^2 + (\Im F(x + ih))^2} dt. \quad (5.7)$$

Proof. The function F maps the upper half-plane into the lower half-plane. Compose harmonic measure of S_0 there with F . The resulting bounded harmonic function on the upper half-plane has the indicated boundary values almost everywhere. The exceptional boundary points are poles of F and preimages of the endpoints of S_0 , a finite set. Its Poisson representation is (5.7). $\square$

6 The almost-everywhere bound

Proof of Theorem 1(ii). The failure set is the union, over rational $0 < c < c_0$ and integers $N \geq 2$, of the measurable sets

$$\{y \in (-1, 1) : \lambda_n(y) \leq c \log n \text{ for every } n \geq N\}.$$

If the assertion fails, one of these sets E has positive measure. For its fixed c, N , we therefore have

$$\lambda_n(y) \leq c \log n \quad (y \in E, n \geq N). \quad (6.1)$$

Choose constants

$$K > 1, \quad 0 < b < a < 1, \quad \frac{2b}{\pi c K} > 1,$$

which is possible because $c < 2/\pi$. Set

$$h = n^{-a}, \quad \eta = n^{-b}, \quad H^\eta(x) = \frac{2}{\pi} \int_\eta^1 \gamma_s(x) \frac{ds}{s}.$$

By (5.3), $H^\eta \leq H$. For $s > 0$ and real x, y, t , the triangle inequality gives

$$|y + is - t| \leq |x + is - t| + |x - y| \leq \left(1 + \frac{|x - y|}{s}\right) |x + is - t|.$$

Taking reciprocal squares and summing against the positive masses yields $\gamma_s(x) \leq (1 + |x - y|/s)^2 \gamma_s(y)$. Integrating over $\eta \leq s \leq 1$, we obtain

$$H^\eta(x) \leq \left(1 + \frac{|x - y|}{\eta}\right)^2 H^\eta(y). \quad (6.2)$$

Fix $x \in E$, and write

$$u = \Re m(x + ih), \quad \gamma = \gamma_h(x), \quad S_x = [u - K\gamma, u + K\gamma].$$

By (5.6) and (5.7), each of the two events

$$m(y) + g(y) \in S_x, \quad m(y) - g(y) \in S_x$$

has probability

$$p_K + o(1), \quad p_K = \frac{2}{\pi} \arctan K > \frac{1}{2},$$

under the Poisson probability $p_h(x-y) dy$. The error is uniform in $x \in E$: for $F = m \pm g$, (5.6) gives $\Re F(x+ih) = u + o(\gamma)$ and $-\Im F(x+ih) = \gamma(1 + o(1))$. Substituting $t = u + \gamma s$ in (5.7) therefore gives $\pi^{-1} \int_{-K}^K (1+s^2)^{-1} ds + o(1) = p_K + o(1)$. The two events have intersection probability at least

$$q_K + o(1), \quad q_K = 2p_K - 1 > 0. \quad (6.3)$$

Choose R so that the Poisson mass of $|y-x| > Rh$ is less than $q_K/8$, and let

$$E_n = \{x \in E : (p_h * \mathbf{1}_{\mathbb{R} \setminus E})(x) < q_K/8\}.$$

The approximate identity theorem gives $|E \setminus E_n| \rightarrow 0$. For all sufficiently large n , (6.3) is greater than $q_K/2$. For $x \in E_n$, the two excluded sets $\mathbb{R} \setminus E$ and $\{|y-x| > Rh\}$ have total Poisson mass less than $q_K/4$. Hence there exists $y \in E$ with $|y-x| \leq Rh$ and both events true. We may take y away from the nodes. Since $m(y) \pm g(y)$ both lie in an interval of length $2K\gamma_h(x)$,

$$|g(y)| \leq K\gamma_h(x).$$

Equations (5.3) and (6.1) give

$$H^\eta(y) \leq H(y) = \lambda_n(y)|g(y)| \leq cK(\log n)\gamma_h(x).$$

Together with (6.2), this yields

$$H^\eta(x) \leq (1 + Rh/\eta)^2 H^\eta(y) \leq cK(1 + Rh/\eta)^2 (\log n)\gamma_h(x) \quad (x \in E_n). \quad (6.4)$$

Let T_n be convolution with the symmetric probability kernel

$$k_n(t) = \frac{1}{b \log n} \int_\eta^1 p_{s-h}(t) \frac{ds}{s}.$$

Its integral is one, since $\int_\eta^1 ds/s = b \log n$. Equation (5.4) then shows that

$$H^\eta = \frac{2b}{\pi} (\log n) T_n \gamma_h.$$

Thus (6.4) becomes

$$\frac{T_n \gamma_h(x)}{\gamma_h(x)} \leq \frac{1}{A_n} \quad (x \in E_n), \quad A_n = \frac{2b}{\pi cK} (1 + Rh/\eta)^{-2} \longrightarrow A > 1. \quad (6.5)$$

Integrate (6.5) over E_n , discard the contribution from $y \notin E_n$, and symmetrize. Positivity of γ_h , symmetry of k_n , and $(r + r^{-1})/2 \geq 1$ give

$$\frac{|E_n|}{A_n} \geq \iint_{E_n \times E_n} k_n(x-y) \frac{\gamma_h(y)}{\gamma_h(x)} dy dx \geq \iint_{E_n \times E_n} k_n(x-y) dy dx. \quad (6.6)$$

The kernels k_n form an approximate identity. In fact, for every fixed $\delta > 0$,

$$\int_{|t|>\delta} k_n(t) dt \leq \frac{C}{b \log n} \int_\eta^1 \min(1, s/\delta) \frac{ds}{s} \longrightarrow 0.$$

It follows that

$$\iint_{E \times E} k_n(x-y) dy dx \longrightarrow |E|.$$

Replacing E by E_n changes this integral by at most $2|E \setminus E_n|$. Taking limits in (6.6) therefore yields

$$\frac{|E|}{A} \geq |E|,$$

contrary to $|E| > 0$ and $A > 1$. This proves the assertion. $\square$

Corollary. *Let $E \subset (-1, 1)$ be measurable with $|E| > 0$, and let $0 < c < 2/\pi$. For all sufficiently large n , there is $x \in E$ such that $\lambda_n(x) > c \log n$.*

Proof. Otherwise $\lambda_n(y) \leq c \log n$ for every $y \in E$ and every n in an infinite set S . The argument above applies to these rows, with all limits taken as $n \rightarrow \infty$ through S , and gives the same contradiction $|E|/A \geq |E|$ with $A > 1$. $\square$

Use of generative AI

GPT-6 Astra generated the proofs and first draft; GPT-5.6 Sol and Claude Opus 5 were used for editorial review. The author reviewed the manuscript. Theorem 1(ii), including the arguments of Sections 5–6, and the set corollary have been formalized in Lean 4 with Mathlib using OpenAI Codex (GPT-6). The corollary is `Erdos1132.eventually_exists_lower_bound`, derived from `uniform_low_set_null` with an upper bound assumed for infinitely many rows. The source and build instructions are at <https://github.com/FireflySentinel/erdos-1132>. Theorem 1(i) is not formalized.

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